Project Report

1. INTRODUCTION

a. Project Overview

This project, "Power BI Inflation Analysis – Journeying Through Global Economic Terrain," analyzes global inflation trends using an interactive Power BI dashboard. It consolidates data from sources like the IMF, World Bank, and OECD, enabling users to track, compare, and forecast inflation. By integrating predictive analytics and visual insights, the project helps policymakers, businesses, and researchers make data-driven economic decisions.

b. Purpose

The project aims to **simplify inflation analysis** by transforming complex economic data into **interactive and accessible insights**. It helps stakeholders understand inflation trends, identify key drivers, and predict future economic conditions. By leveraging **real-time data visualization and forecasting models**, it enhances decision-making for **policymakers, economists, investors, and businesses**, ensuring informed economic planning and stability.

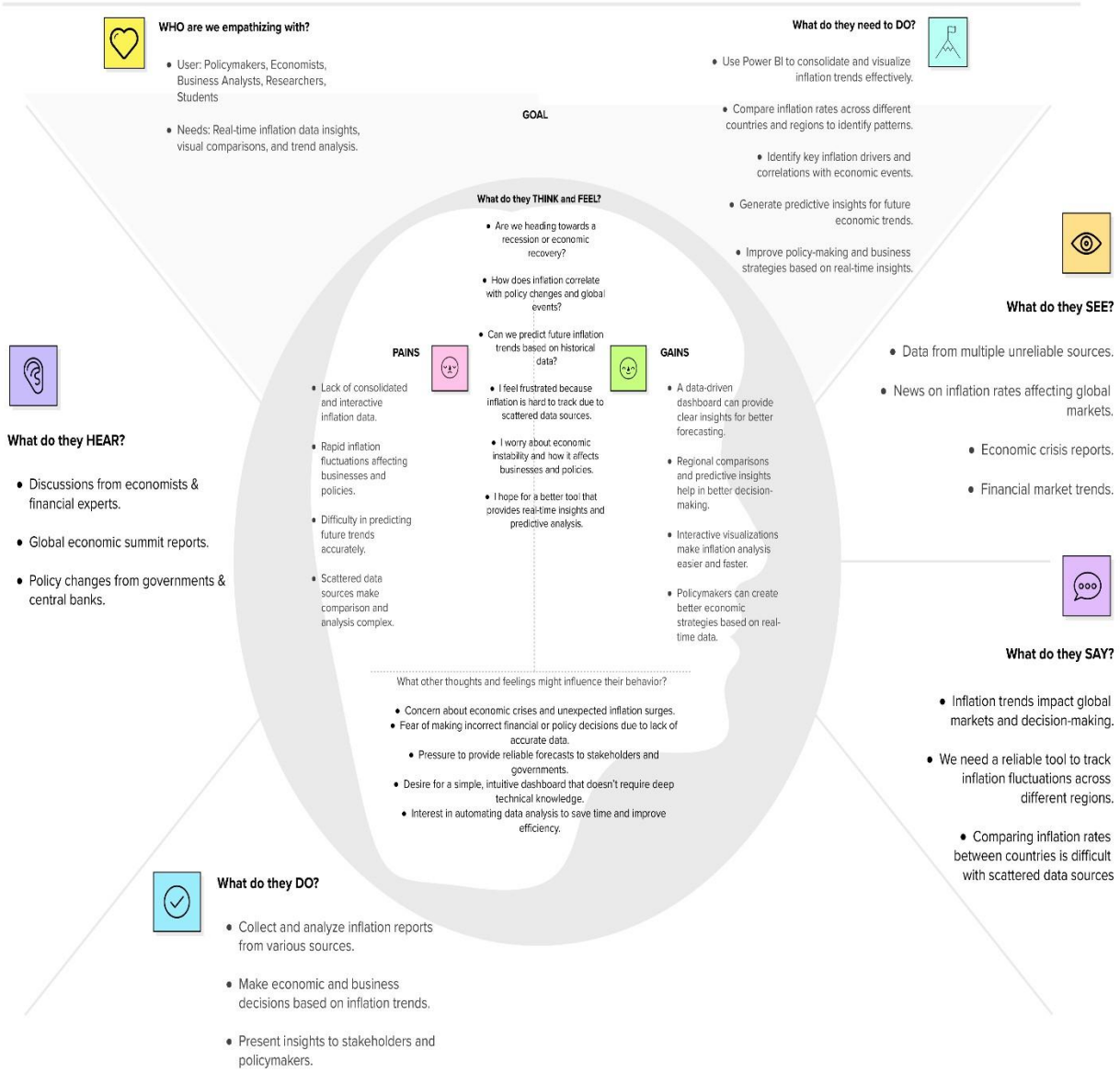
2. <u>IDEATION PHASE</u>

a. Problem Statement

Analyzing inflation trends is difficult due to scattered data and complex reports. This project, **"Power BI Inflation Analysis – Journeying Through Global Economic Terrain,"** develops an interactive Power BI dashboard to consolidate global inflation data, provide real-time insights, and integrate predictive analytics, enabling users to track, compare, and forecast inflation trends effectively.



b. Empathy Map Canvas



3. REQUIREMENT ANALYSIS

a. Customer Journey map



b. Solution Requirement

Solution Requirements (Functional & Non-functional)

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Ingestion	Import global inflation dataset from Kaggle
FR-2	Data Cleaning	Handle missing values, standardize column names, and format data.
FR-3	Data Visualization	Generate Power BI reports (Line Charts, Heatmaps, Bar Graphs)
FR-4	Filtering & Analysis	Provide filtering options (by country, region, year).
FR-5	Predictive Insights	Use machine learning to forecast inflation trends (if applicable).

Non-functional Requirements:

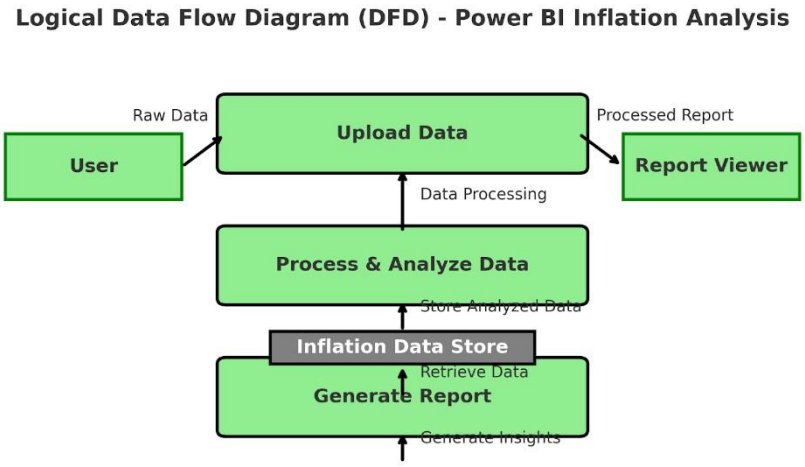
Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should be user-friendly and interactive.
NFR-2	Security	Ensure data integrity and restrict unauthorized modifications.
NFR-3	Reliability	Ensure consistent and accurate results across multiple executions.
NFR-4	Performance	Optimize Power BI queries to ensure fast rendering of reports.
NFR-5	Availability	The system should be accessible 24/7 with minimal downtime.
NFR-6	Scalability	Ensure the system can handle additional datasets in the future.

c. Data Flow Diagram

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.



d. Technology Stack

Technical Architecture:

This document outlines the technology stack used for the Power BI Inflation Analysis project. The system is built on Power BI Desktop and follows a structured data workflow from raw dataset ingestion to visualization and reporting. The entire process is conducted offline, without cloud deployment.

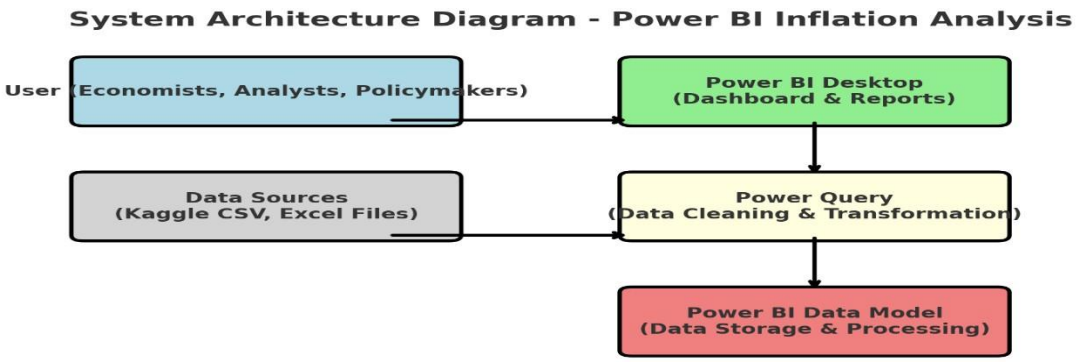


Table-1 : Components & Technologies:

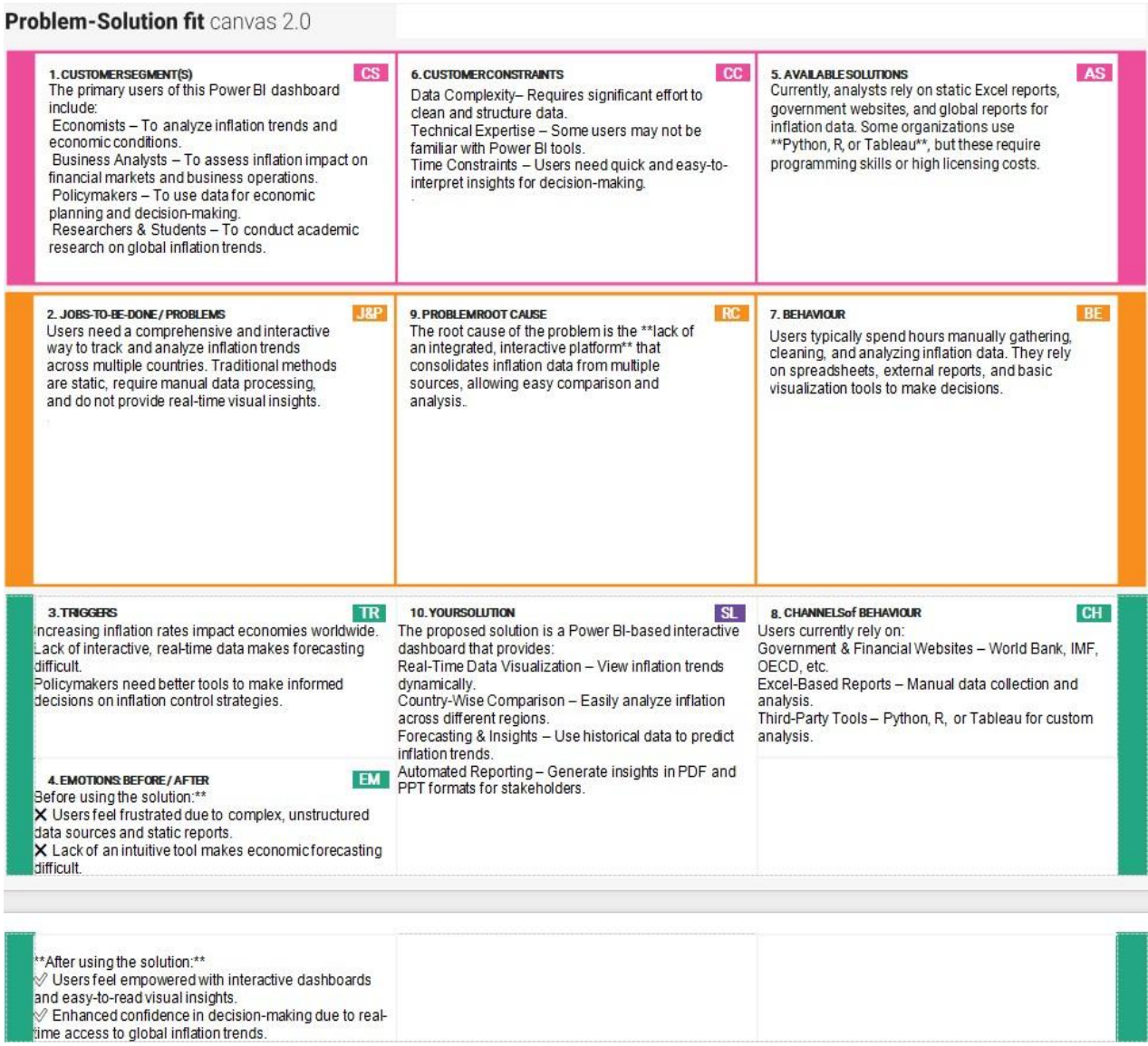
S.No	Component	Description	Technology
1.	User Interface	Power BI Desktop	Allows users to interact with dashboards and reports.
2.	Data Import	CSV/Excel Files (Kaggle Dataset)	Provides raw inflation data for analysis.
3.	Data Processing	Power Query (M Language)	Cleans, transforms, and prepares data.
4.	Data Storage	Power BI Data Model	Stores structured data for visualization.
5.	Visualization	Power BI Charts & Reports	Generates insights through graphs and tables.
6.	Interactivity	Power BI Slicers & Filters	Enables users to explore data dynamically.
7.	Export & Sharing	Power BI PDF, PPT	Allows sharing of reports in standard formats.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Power BI uses opensource connectors for data import.	Power Query, DAX
2.	Security Implementations	Local access control and dataset encryption.	Role-Based Access Control (RBAC)
3.	Scalable Architecture	Handles large datasets with optimized queries.	Power BI Data Model
4.	Availability	Works offline with Power BI Desktop, ensuring constant access.	Local Deployment
5.	Performance	Optimized using DAX calculations and indexing.	Power BI Engine

4. PROJECT DESIGN

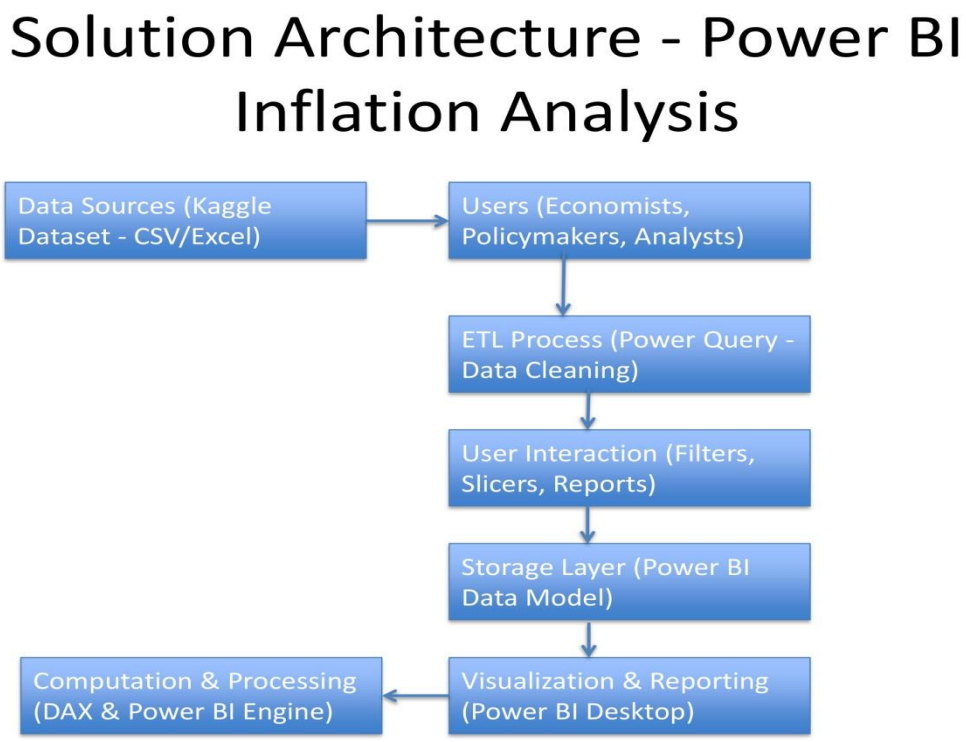
a. Problem Solution Fit



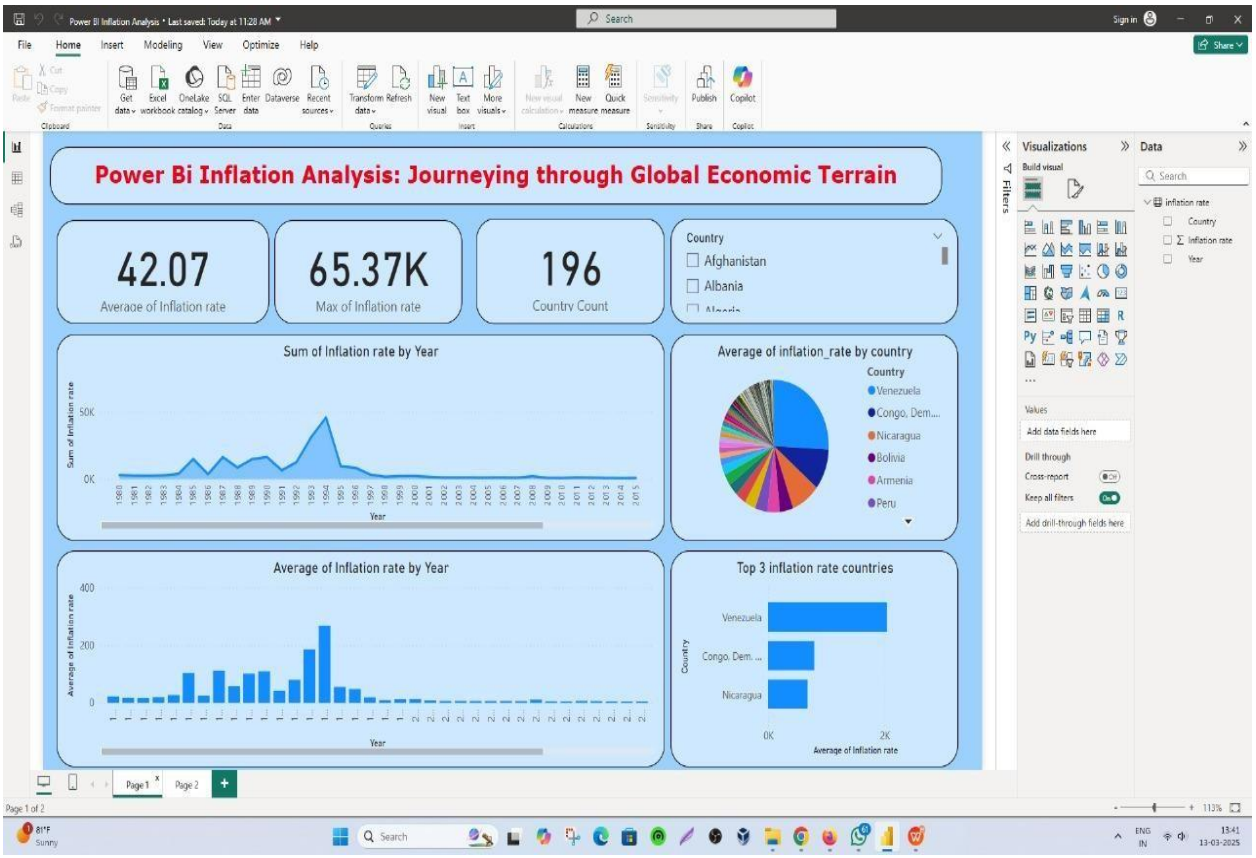
b. Proposed Solution

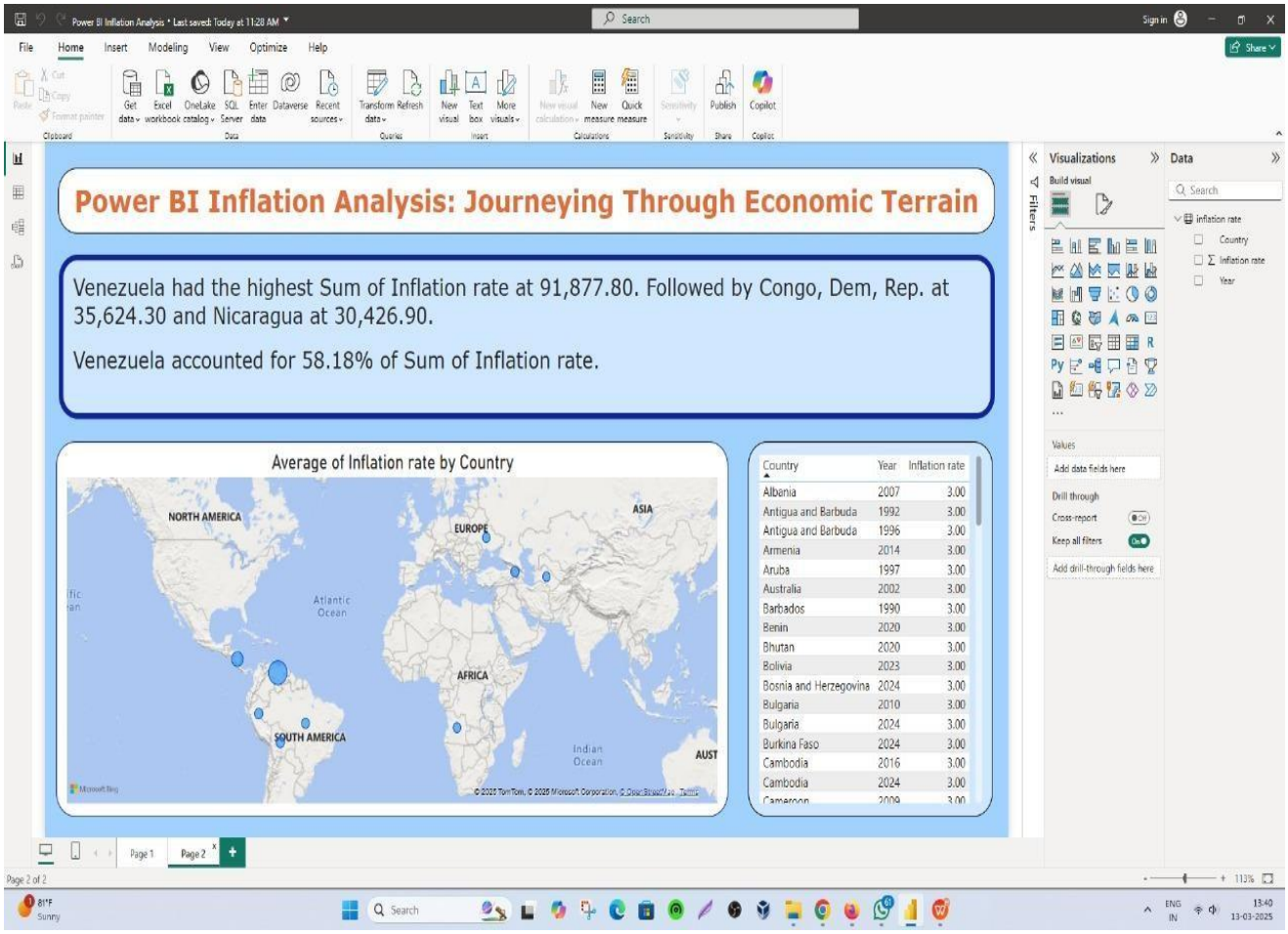
S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Inflation analysis is complex due to scattered data sources, lack of interactive tools, and difficulty in forecasting trends. This project addresses these challenges by providing a centralized, interactive, and predictive Power BI dashboard.
2	Idea / Solution Description	The solution is a Power BI-based interactive dashboard that consolidates global inflation data, provides real-time visual insights, and integrates predictive analytics.
3	Novelty / Uniqueness	Unlike traditional methods (spreadsheets, static reports), this dashboard offers real-time data visualization, AI-powered forecasting, and interactive user controls.
4	Social Impact / Customer Satisfaction	Helps policymakers design better economic policies, enables businesses to make informed pricing and investment decisions, and assists researchers and students in understanding economic trends.
5	Business Model (Revenue Model)	If commercialized, revenue can be generated through subscription-based access, premium features (advanced forecasting, data exports), or collaborations with financial institutions, government bodies, and economic research firms.
6	Scalability of the Solution	The solution can be expanded globally by integrating more datasets, enhancing AIdriven analytics, and adding customized

c. Solution Architecture



6. RESULTS a. Output Screenshots


Power Bi Inflation Analysis: Journeying through Global Economic Terrain
42.07
Averaae of Inflation rate
65.37K
Max of Inflation rate
196
Country Count
Country
Afghanistan
Albania
Sum of Inflation rate by Year
Sum of Inflation rate
50K
0K
Year
Average of inflation_rate by country
Country
Venezuela
Congo, Dem....
Nicaragua
Bolivia
Armenia
Peru
Average of Inflation rate by Year
Average of Inflation rate
400
200
0
Year
Top 3 inflation rate countries
Country
Venezuela
Congo, Dem. ...
Nicaragua
0K
2K
Average of Inflation rate
Visualizations
Data
Filters
Page 1
Page 2



8. ADVANTAGES & DISADVANTAGES

Advantages-

- **Centralized Data Integration** – The project consolidates inflation data from multiple sources like the World Bank, IMF, and OECD, providing a unified platform for analysis.
- **Interactive Visualizations** – Power BI's dynamic charts, heatmaps, and dashboards make complex inflation data more comprehensible.
- **Predictive Analytics** – AI-driven forecasting models help in predicting future inflation trends, assisting in better economic planning.
- **Real-Time Data Analysis** – Users can analyze inflation trends as data updates occur, making insights more relevant and timely.
- **Enhanced Decision-Making** – Policymakers, businesses, and researchers can make datadriven decisions using customized reports and comparative analysis.
- **User-Friendly Interface** – Power BI's drag-and-drop functionality and interactive filters improve accessibility, even for non-technical users.
- **Scalability** – The solution can be expanded globally, integrating more datasets and AI-driven insight.

Disadvantages-g

Data Dependency – The accuracy of insights depends on the quality and availability of data from external sources.

Technical Expertise Required – Users need knowledge of Power BI, DAX, and data modeling to create and customize dashboards.

Limited Real-Time Updates – If data is not linked to live APIs, manual updates are required for the latest information.

Performance Issues with Large Datasets – Processing extensive datasets in Power BI may slow down dashboard performance.

Limited Offline Access – Power BI dashboards require an internet connection for updates and cloud-based features.

Cost Considerations – Advanced features in Power BI Pro or Power BI Premium require paid licenses, which may not be affordable for all users.

9. CONCLUSION

The **Power BI Inflation Analysis** project simplifies inflation trend analysis by integrating global data into an interactive dashboard. It enhances **decision-making** through real-time insights and predictive analytics. Despite challenges like data dependency, its **scalability and usability** make it a valuable tool for policymakers, businesses, and researchers in understanding economic trends.

10. FUTURE SCOPE

The future scope of this project includes automated data integration, AI-driven forecasting, and expanded economic indicators for deeper analysis. Enhancements like cloud deployment, mobile accessibility, and customizable reports will improve usability. Integrating real-time alerts and AI-powered insights can further assist policymakers, businesses, and researchers in economic planning. Scalability across industries will make it a comprehensive, data-driven tool for global inflation analysis.

11. APPENDIX

Dataset Link

Global Inflation Dataset

GitHub & Project Demo Link https://github.com/shreeyamane/Power-BI-Inflation-Analysis-Journeying-Through-GlobalEconomic-Terrain.